

DEEPSIFT

Measured downlink autonomy for bandwidth-constrained missions · independent research prototype · v1 (deepsift-v1-research)

PROBLEM. Rover imagery competes for limited onboard storage and relay downlink. Something has to decide what is sent at full quality, what goes down as a thumbnail, and what waits.

QUESTION. How far can archived rover traverse imagery be reduced while preserving spatial mission coverage and stereo integrity?

DATA. Public NASA PDS archives: Curiosity Navcam raw EDR (stereo, all product tiers, ICER/LOCO label metadata) and PLACES rover localizations. Downlink cost comes from label estimates (lines × samples × compression rate / 8).

EXPERIMENT.

stage	sols	purpose
Development	412–430	method design and tuning
Validation 1	779–820	first out-of-sample check
Validation 2	1100–1129	fresh check after simplification
Final held-out test	950–979	one run

Every configuration and pass/fail rule was committed to git before the data that tested it was downloaded. The four periods are never pooled.

FINAL HELD-OUT RESULT — Curiosity Navcam sols 950–979: 12 traverses, 679 archived frames. Method: Scheduler V3 + rover-position sampling at 1/4 **traverse-frame retention**. Pre-registered criterion: **PASS**.

full-quality traverse bytes	5 m spatial coverage	max distance, archived frame → retained frame	broken stereo pairs
26.1 %	100 %	2.05 m	0

WHAT GENERALIZED

- **Rover-position sampling:** farthest-point selection on PLACES positions. 5 m coverage was 1.000 in all four periods.
- **Scheduler V3:** stereo-safe and progressive. It sends thumbnail pairs, then compressed pairs, then full pairs, and never one eye alone. It had 0 monotonicity violations in all four periods.

WHAT DID NOT EARN A PRIMARY ROLE

- **Jev semantic ranking:** AUROC vs rules +0.006 (95 % CI −0.020 to +0.037); stopped by a pre-registered rule.
- **Telemetry image ranking:** image-independent by construction; weakest coverage of six strategies.
- **pHash representative selection:** constrained false-merge rate 0.15–0.19 on validation; at most 1.03× compression.
- **QUALITY_V2:** false-positive rate 3.6 % in development and 10.6 % on validation.
- **Embedding-assisted selection:** visual-change gain over position +0.044 in development, then +0.004, +0.001 and +0.005, against a +0.020 threshold. No measurable added value.

LIMITATIONS

- **Retrospective.** The PDS contains only observations that were actually downlinked, so this is re-prioritization of archived data, not reconstruction of the onboard stream.
- **Simulated compressed tier.** The compressed stereo tier is simulated.
- **Interpolated positions.** Positions come from PLACES, mostly the nearest pose within a drive.
- **Narrow scope.** One rover, one camera, traverse sequences only.
- **Proxy metrics.** Coverage is geometric, not a measure of science value.
- **Not flight-ready, and not reviewed, used or validated by NASA/JPL.**

LINKS

Demo	{{DEMO_URL}} (/final-test)
GitHub	{{GITHUB_URL}}
Paper	docs/deepsift-paper.md · artifacts/public/deepsift-v1-paper.pdf
Reproducibility	{{DEMO_URL}}/reproducibility · scripts/release_integrity.py --verify